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Pharmacokinetics

(Drug Absorption)

BY

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Objectives:

1. Discuss mechanisms of absorption of drugs from the GI tract
2. Determine factors influencing drug absorption
3. Discuss drug bioavailability

ABSORPTION OF DRUGS:

- ✚ Absorption is the **transfer of a drug from site of administration** to the **blood stream** via several mechanisms.
- ✚ For **IV administration**, **absorption is complete**, or the **total dose of administered drug reaches the systemic circulation (100% bioavailability)**. Drug delivery by **other routes** may result in only **partial absorption** and, thus, **lower bioavailability**.

Mechanisms of absorption of drugs from the GI tract:

1- Passive diffusion:

- ❖ **Water-soluble drugs** penetrate the cell membrane (**CM**) through **aqueous channels or pores**, whereas **lipid-soluble drugs** readily cross **CM** due to their **solubility in the membrane lipid bilayers**

2- Facilitated diffusion:

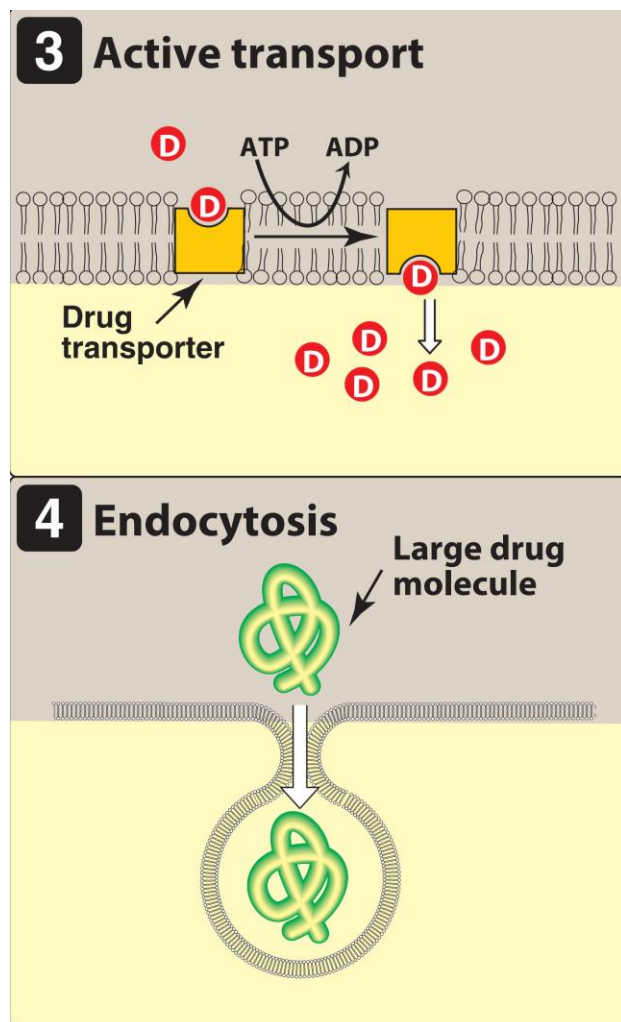
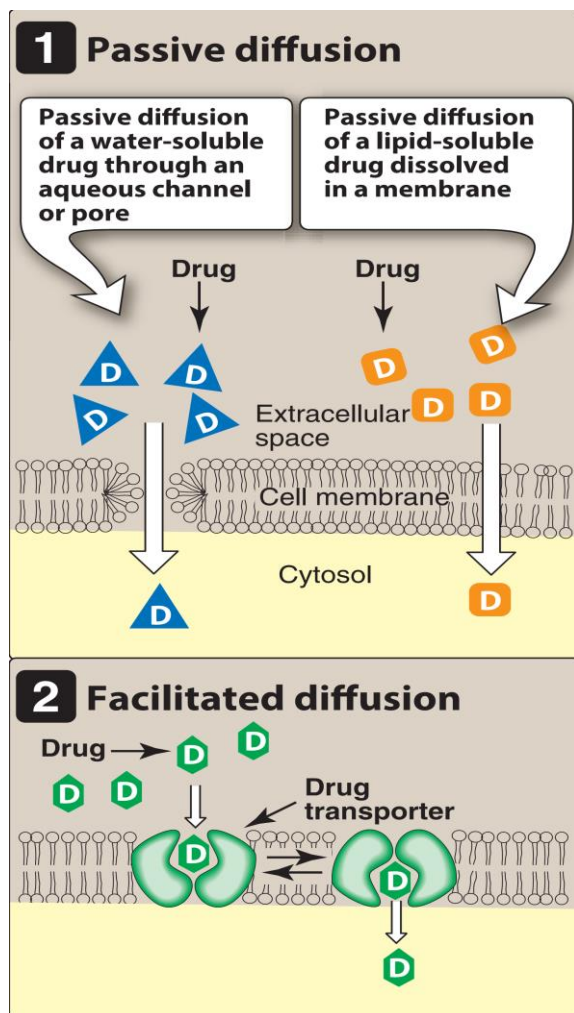
- ❖ Occur through **specialized transmembrane carrier proteins** that facilitate the passage of large molecules from an **area of high concentration to an area of low concentration**.
- ❖ It **does not require energy**

3- Active transport:

- ❖ **Energy-dependent active transport** is driven by the hydrolysis of ATP moving drugs **against a concentration gradient**, from a **region of low drug concentration to one of higher concentration**.

4- Endocytosis and exocytosis:

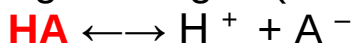
- ❖ Used to transport drugs of **exceptionally large size** across the cell membrane
- ❖ **Exocytosis** is the reverse of **endocytosis** and is used by cells to secrete many substances by a similar vesicle formation process.
- ❖ **Vitamin B12** is **transported across the gut wall by endocytosis**, whereas certain neurotransmitters (e.g., **noradrenaline**) are stored in intracellular membrane-bound vesicles in the nerve terminal and are **released by exocytosis**.



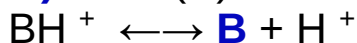
Factors influencing absorption:

1- Effect of pH on drug absorption:

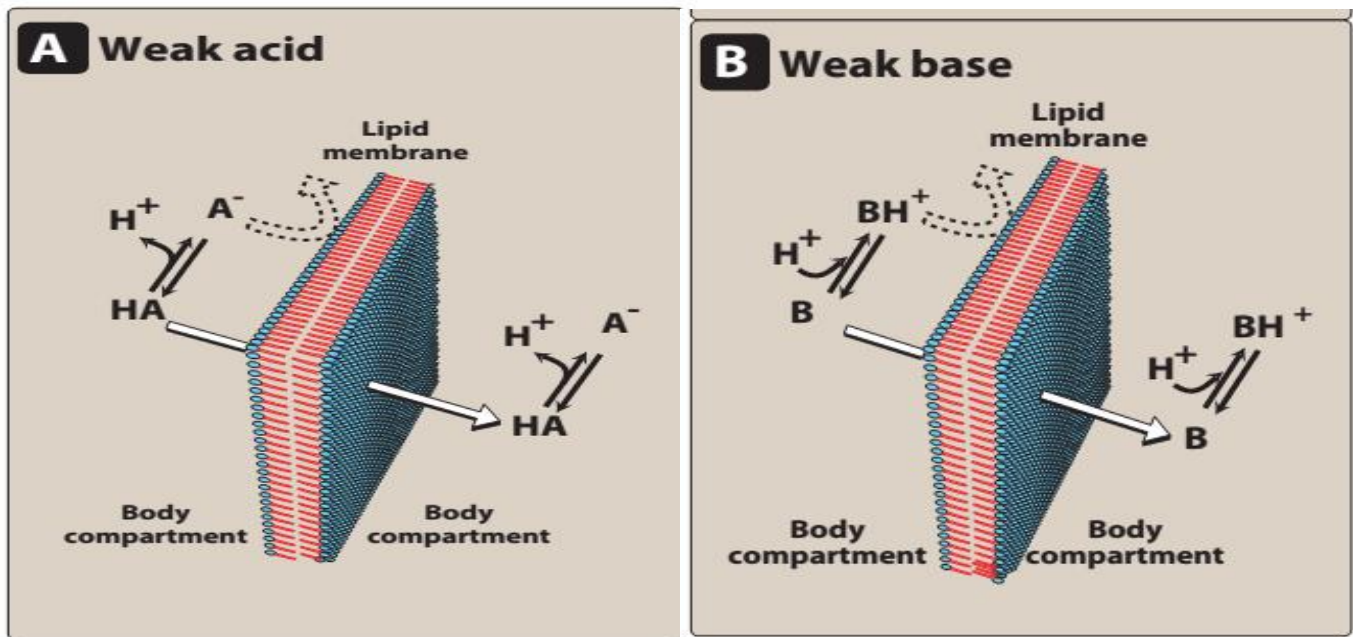
Most drugs are either **weak acids** or **weak bases**. **Acidic drugs (HA)** release a proton (H^+), causing a charged (ionized) anion (A^-) to form:



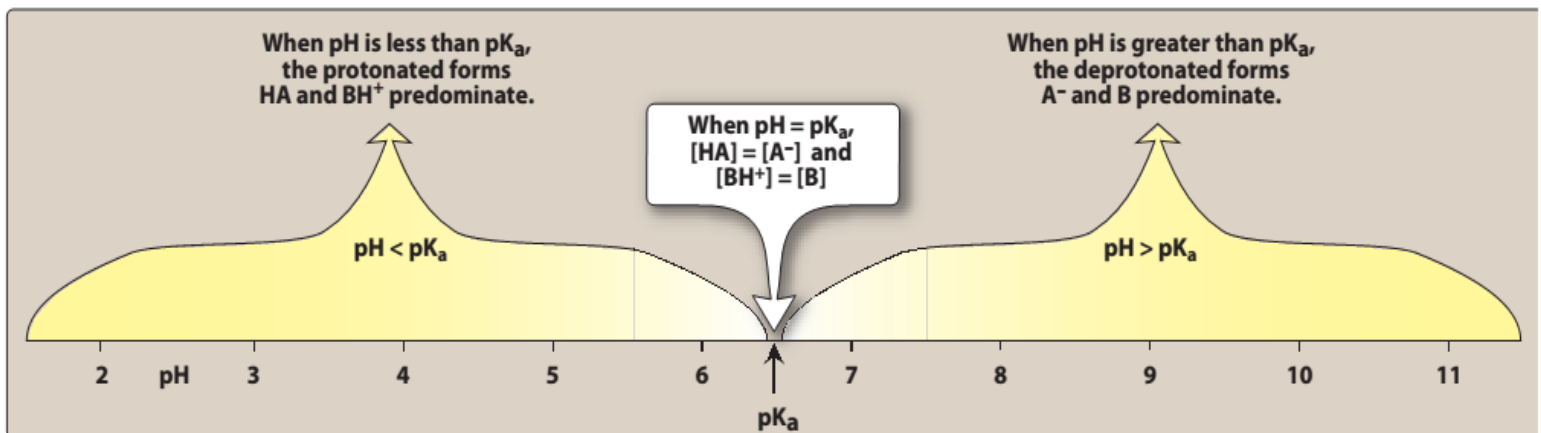
Weak bases (BH^+) can also release an H^+ . However, the protonated form of basic drugs is usually charged, and loss of a proton produces the **uncharged (non-ionized)** base (**B**):



A drug passes through membranes more readily if it is uncharged (non-ionized). Thus, for a **weak acid**, the **uncharged, protonated HA can permeate through membranes**, and **A^- cannot**. For a **weak base**, the **uncharged form ,B , penetrates through the cell membrane**, but **BH^+ , the protonated form, does not**.



For each drug (acid or base), the **ratio between charged and uncharged forms** is determined by the **pH at the site of absorption** and by **the strength of the weak acid or base**, which is represented by the ionization constant, **pKa**. The **pKa** is a measure of the strength of the interaction of a compound with a proton [The lower the pKa of a drug, the **more acidic** it is. Conversely, the **higher the pKa**, the **more basic** is the drug].



In acidic media (low PH, e.g. stomach), **acidic drugs will be non-ionized (HA) or lipid soluble**, while **basic drugs will be ionized (BH⁺) or water soluble**.

In basic media (high PH, e.g. small intestine), **acidic drugs will be ionized (A⁻) or water soluble**, while **basic drugs will be non-ionized (B) or lipid soluble**.

2- Blood flow to the absorption site:

Because **blood flow to the intestine** is much greater than the flow to the stomach, absorption from the intestine is favored over that from the stomach.

3- Total surface area available for absorption:

Intestine has much greater surface area than the stomach

4- Contact time at the absorption surface

- If a drug moves through the GIT very quickly, as can happen with **severe diarrhea**, it is not well absorbed.
- The presence of **food in the stomach** both **dilutes the drug** and **slows gastric emptying**. Therefore, a drug taken with a meal is generally absorbed more slowly.

5- Expression of P-glycoprotein:

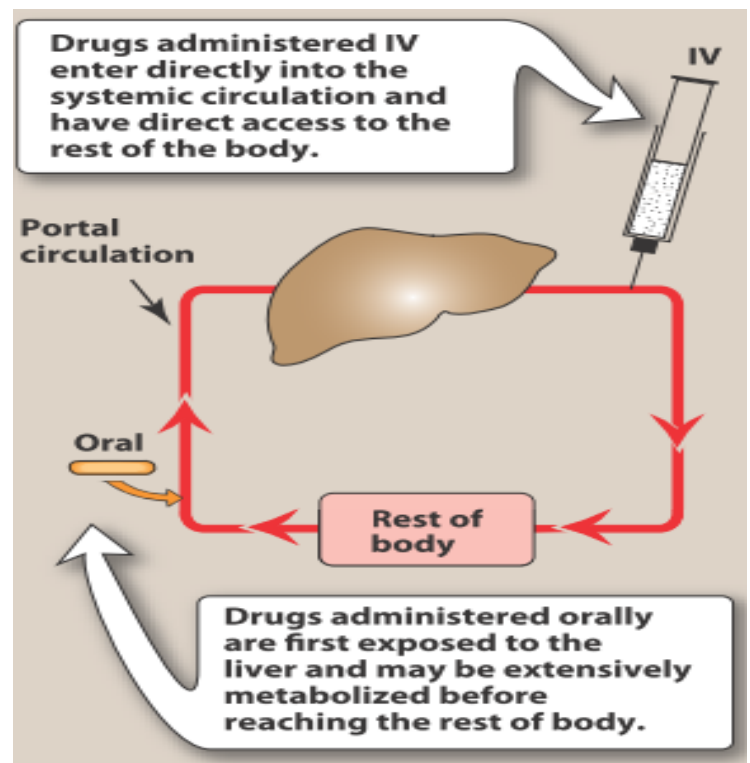
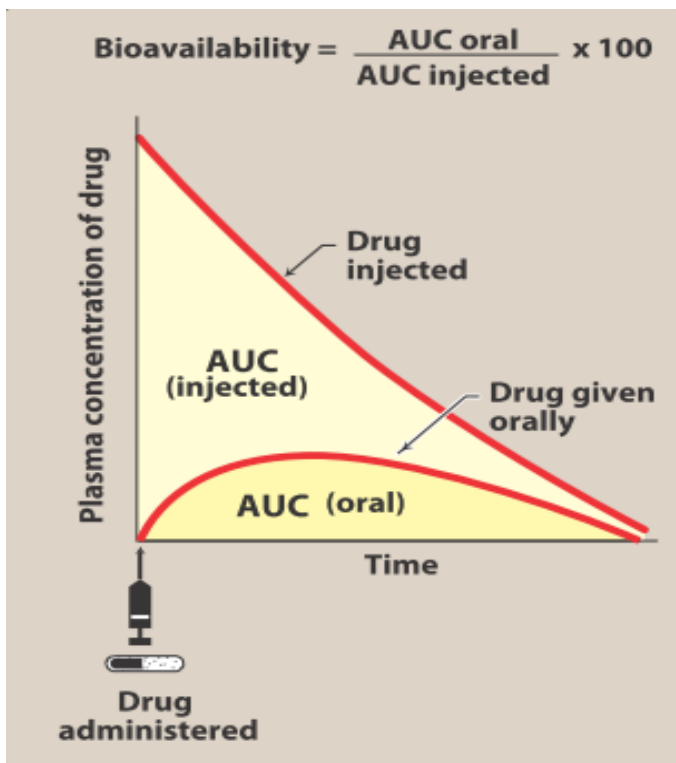
P-glycoprotein is a **multidrug transmembrane transporter protein** responsible for transporting various molecules, including drugs, across cell membranes. It is expressed throughout the body, and **its functions include:**

- In the **liver**: **transporting drugs into bile** (for elimination)
- In **kidneys**: **pumping drugs into urine** (for excretion)
- In the **placenta**: **transporting drugs back into maternal blood**
- In the **intestines**: **transporting drugs into the intestinal lumen** (reducing drug absorption into the blood)
- In the **brain capillaries**: **pumping drugs back into blood** (limiting drug access to the brain)

Thus, in areas of high expression, P-glycoprotein reduces drug absorption.

Bioavailability

- It is the fraction of administered drug that reaches the systemic circulation.
- It is important for calculating drug dosages for enteral routes of administration.



Factors that influence bioavailability:

- 1- **First-pass hepatic metabolism:** drug is absorbed across the GI tract; it first enters the portal circulation before entering the systemic circulation. If the drug is **rapidly metabolized** in the liver or gut wall during this initial passage. (**high first pass metabolism for a drug = low oral bioavailability**)
- 2- **Solubility of the drug:** **hydrophilic** drugs are **poorly absorbed** because of their inability to cross the lipid-rich Cell Membrane
- 3- **Chemical instability:** Some drugs, such as penicillin **G**, are **unstable in the pH of the gastric contents**. Others, such as insulin, are destroyed in the GIT.
- 4- **Nature of the drug formulation:** For example, **particle size**, **salt form**, **crystal polymorphism**, **enteric coatings**, and the presence of **excipients**.

Bioequivalence:

Two related drug preparations are bioequivalent if they show **comparable bioavailability** and **similar times to achieve peak blood concentrations**.